

MrDresser

Software Design Document

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1.0 INTRODUCTION

1.1 Purpose

This SWDD describes the system architecture, component design, and data organization of MrDresser – a smart outfit recommendation system using biometric and contextual data. It is intended for developers, testers, and system architects.

1.2 Scope

MrDresser is a mobile/web application offering personalized outfit suggestions based on face shape, skin tone, weather, event, and user profile data. It aims to streamline the outfit selection process with ML-powered recommendations.

1.3 Overview

This document includes system architecture, module design, data schemas, component descriptions, and interface layouts required for MrDresser implementation.

1.4 Reference Material

- IEEE 830-1998
- OpenWeatherMap API
- Google Maps API
- SRS Document

1.5 Definitions and Acronyms

- API: Application Programming Interface
- ML: Machine Learning
- UI: User Interface
- DFD: Data Flow Diagram

2.0 SYSTEM OVERVIEW

MrDresser provides a smart fashion experience using a user's biometric profile, environmental conditions (like weather), and event context to recommend clothing. The system uses AI-based image processing, real-time APIs, and vendor integration.

3.0 SYSTEM ARCHITECTURE

3.1 Architectural Design

The system is modular, consisting of the following major subsystems:

- **User Management:** Handles user profile creation, authentication.
- **ML Engine:** Performs face detection, skin tone analysis, and recommendation logic.
- **Vendor Module:** Manages vendor profiles, product listings.
- **Search & Location:** Enables product filtering and map integration.
- **Admin Dashboard:** Handles content moderation and approval.

3.2 Decomposition Description

Object-Oriented Description:

- **User:** Attributes (name, age, gender, profile image), Methods (editProfile, getRecommendations)
- **Vendor:** Attributes (shopName, productList), Methods (uploadProduct, editShop)
- **Admin:** Methods (approveVendor, deleteProduct)
- **MLService:** Methods (detectFaceShape(), recommendOutfit())
 - **Sequence Diagram:**
- User submits image → MLService processes → Recommendations returned
- Vendor adds product → Validated → Admin approval → Published

3.3 Design Rationale

The modular OO design was selected for scalability and maintenance. External APIs were chosen for reliability (OpenWeatherMap, Google Maps). MongoDB was selected for its flexibility with user data and image storage.

4.0 DATA DESIGN

4.1 Data Description

- **User Data:** JSON documents in MongoDB with biometric and contextual fields.
- **Vendor Data:** Product metadata and media files.
- **Recommendation Data:** Stored temporarily and refreshed on demand.

4.2 Data Dictionary

Entity	Attributes
User	userID, name, age, weight, image, gender
Vendor	vendorID, shopName, location, productList
Product	productID, name, category, price, image
AdminLog	logID, actionType, timestamp

5.0 COMPONENT DESIGN

Each function (OO):

- User::getRecommendations()
 - **Input:** profileData, weatherData, event
 - **Process:** analyzeFace(profileImage), matchClothes(skinTone, weather, event)
 - **Output:** List of outfits
- Vendor::uploadProduct()
 - **Input:** productDetails
 - **Process:** validate(), storeInDB(), notifyAdmin()
- Admin::approveVendor()
 - **Input:** vendorID
 - **Process:** updateStatus("approved"), logAction()

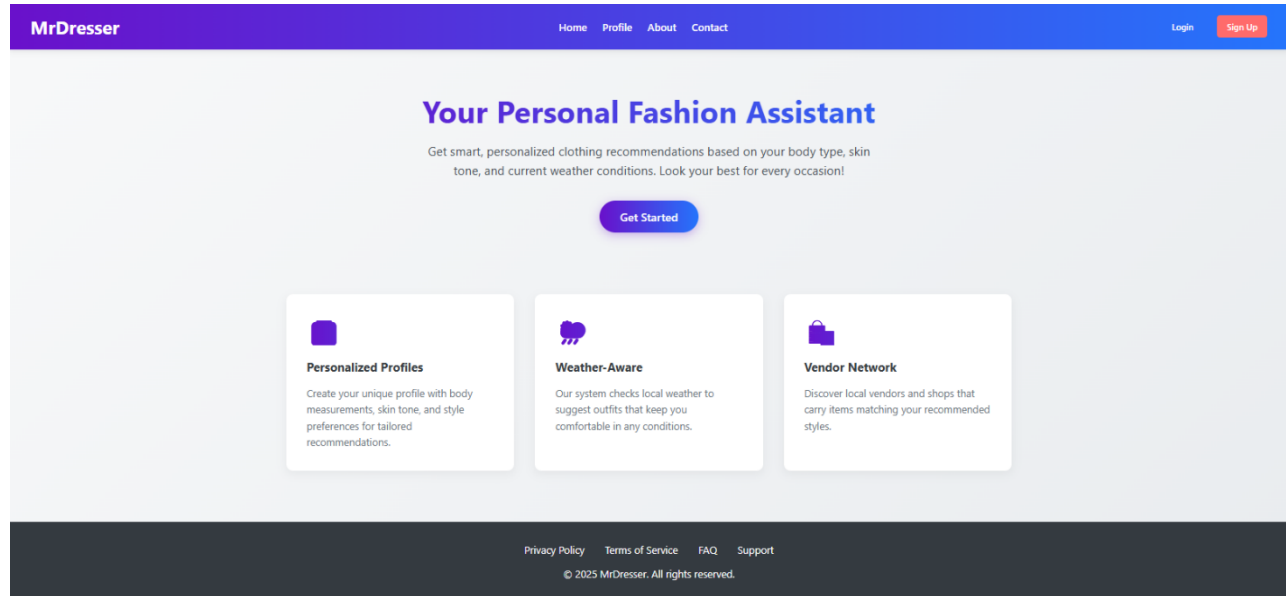
6.0 HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

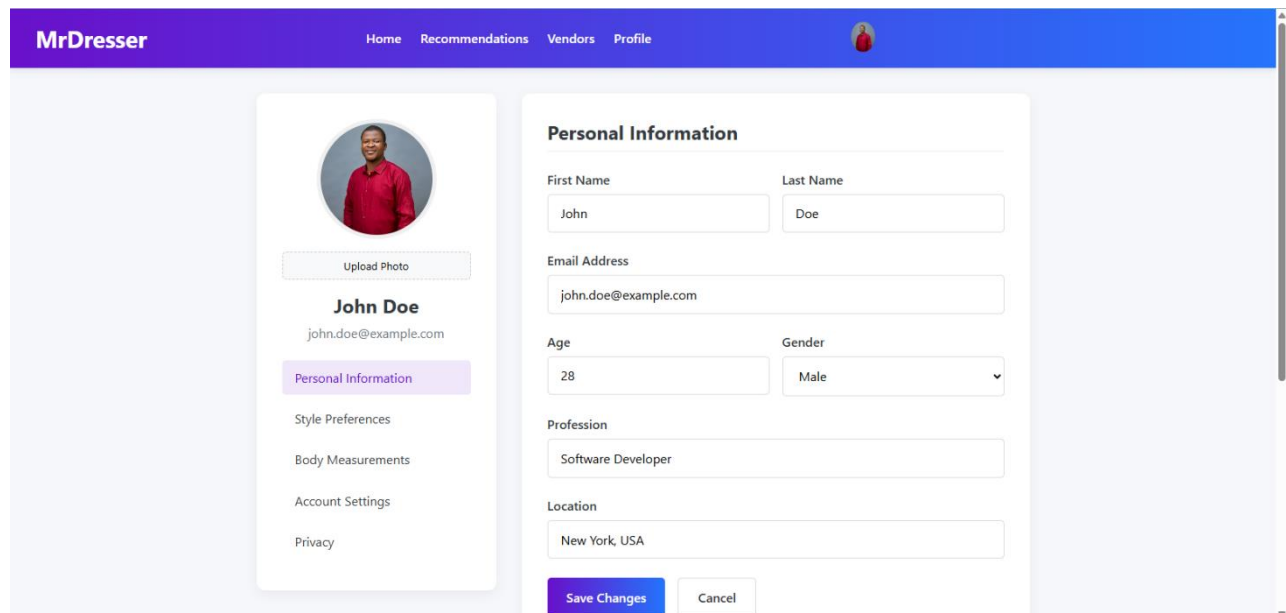
- **User Panel:** Profile editor, photo upload, recommendation viewer
- **Vendor Panel:** Shop creation, product upload
- **Admin Panel:** Content review, analytics dashboard

6.2 Screen Images

Home screen



Profile editor



Biometric Analysis

Upload a clear photo of your face to enable automatic detection of face shape and skin tone for better recommendations.

Face Photo

No file chosen

Face Shape
Oval

Skin Tone
Medium Warm

Color Palette
Autumn

Analyze Photo

Product recommendation results

MrDresser

HomeRecommendationsVendorsProfile

User

Your Personalized Recommendations

Filter & Sort

Recommended for: Business Casual at Work (68°F / 20°C, Sunny)

Based on: Your oval face shape, medium warm skin tone, and preference for classic styles

Color Palette: Autumn colors (Earthy tones, warm hues)

Top Picks For You

Classic Business Casual
A timeless combination of navy blazer, light blue shirt, and khaki chinos perfect for your workplace.

Smart Casual Ensemble
Dark denim jeans with a crisp white shirt and brown leather shoes for a polished yet relaxed look.

Formal Office Look
Charcoal gray suit with a light pink dress shirt and burgundy tie for important meetings.

Complete Outfits

Weekend Casual
Comfortable yet stylish with olive green chinos, striped tee, and white sneakers.

Casual \$129.99

View Details

Save

Casual Comfort

Date Night Outfit
Dark jeans, fitted maroon shirt, and leather jacket for a stylish evening look.

Evening \$219.99

View Details

Save

NightStyle

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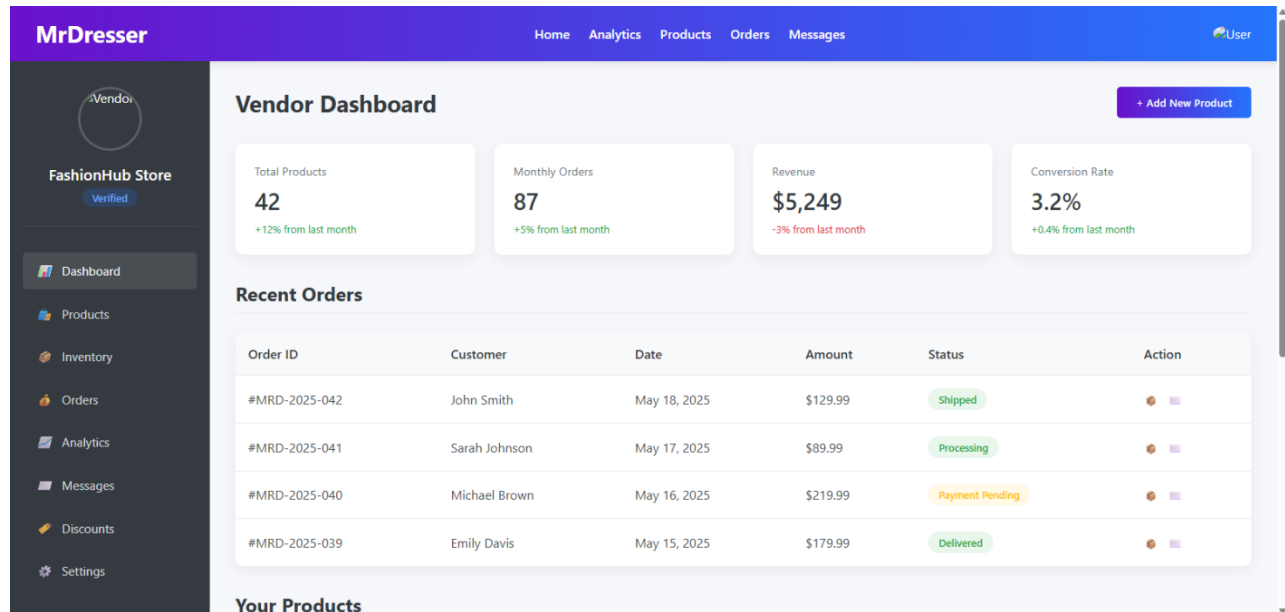
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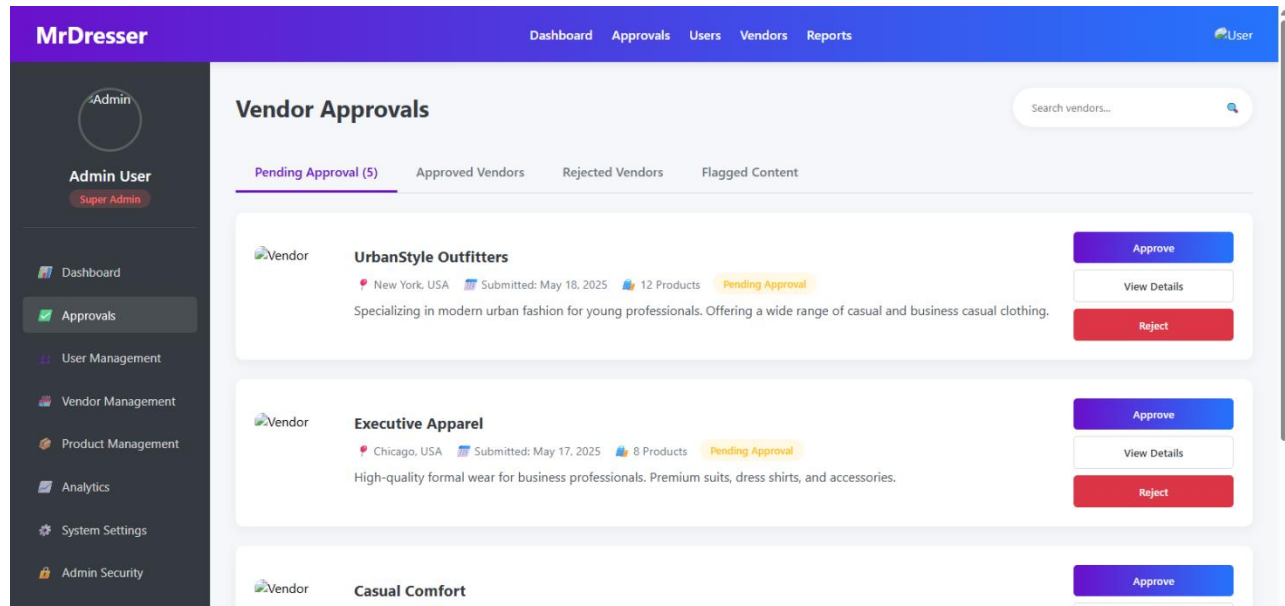
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Vendor dashboard



Admin approval interface



6.3 Screen Objects and Actions

Screen	Objects	Actions
Profile Page	TextFields, UploadBtn	Enter data, upload image
Vendor Dashboard	Form, SubmitBtn	Create profile, add products
Search Page	SearchBar, Filters	Keyword search, filter list
Admin Panel	Table, ActionBtns	Approve, Reject, Delete

7.0 REQUIREMENTS MATRIX

Requirement ID	Description	Component
REQ-1 to REQ-7	User Profile & Personalization	User Module, ML Engine
REQ-6 to REQ-9	Vendor Management	Vendor Module
REQ-10 to REQ-13	Product Search	Search & Location Module
REQ-14 to REQ-17	Location-based Services (optional)	Location Module, Map API
REQ-18 to REQ-21	Admin Dashboard	Admin Module

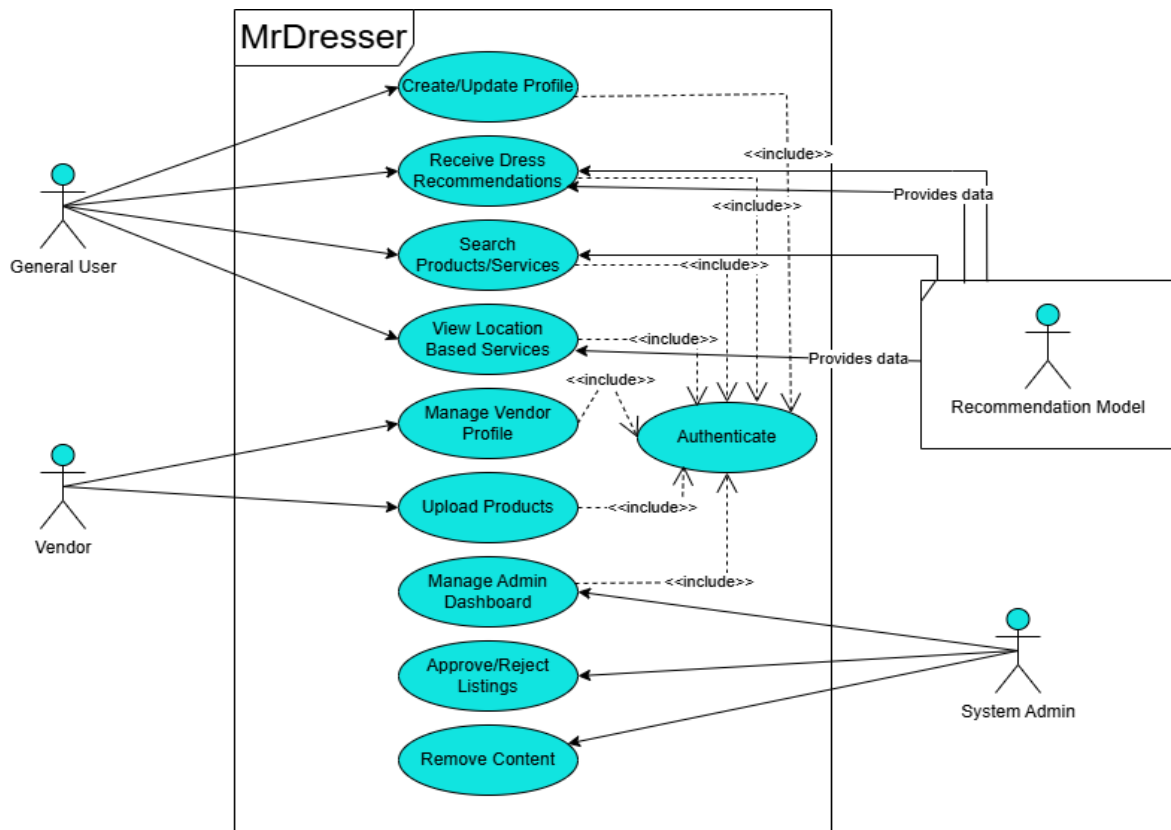
8.0 APPENDICES

Appendix A: Technologies Used

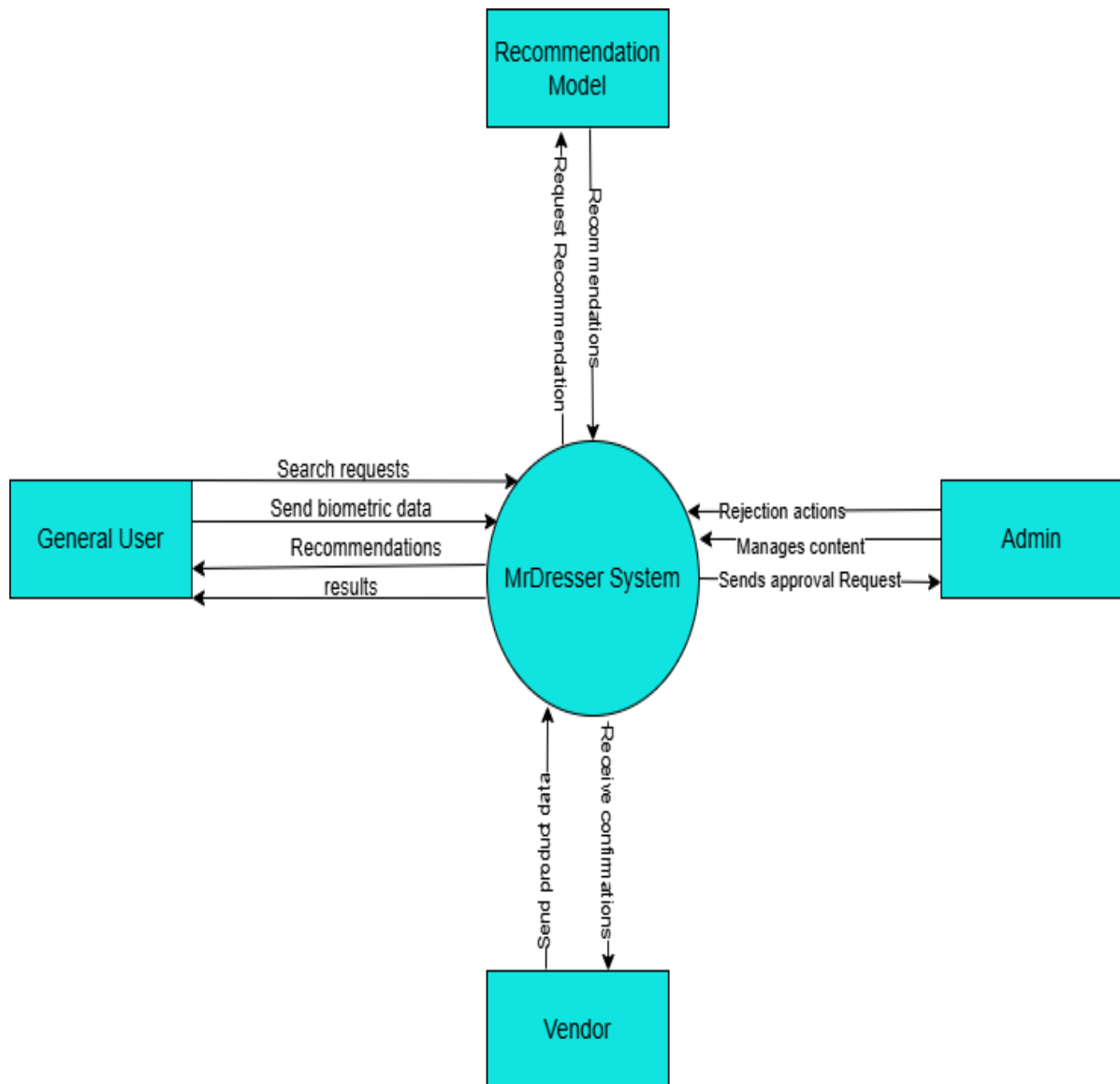
- **Frontend:** React.js, HTML, CSS
- **Backend:** Python (Flask/Django)
- **Database:** MongoDB
- **APIs:** OpenWeatherMap, Google Maps
- **Libraries:** OpenCV, TensorFlow Lite

Appendix B: Diagrams

Use Case Diagram



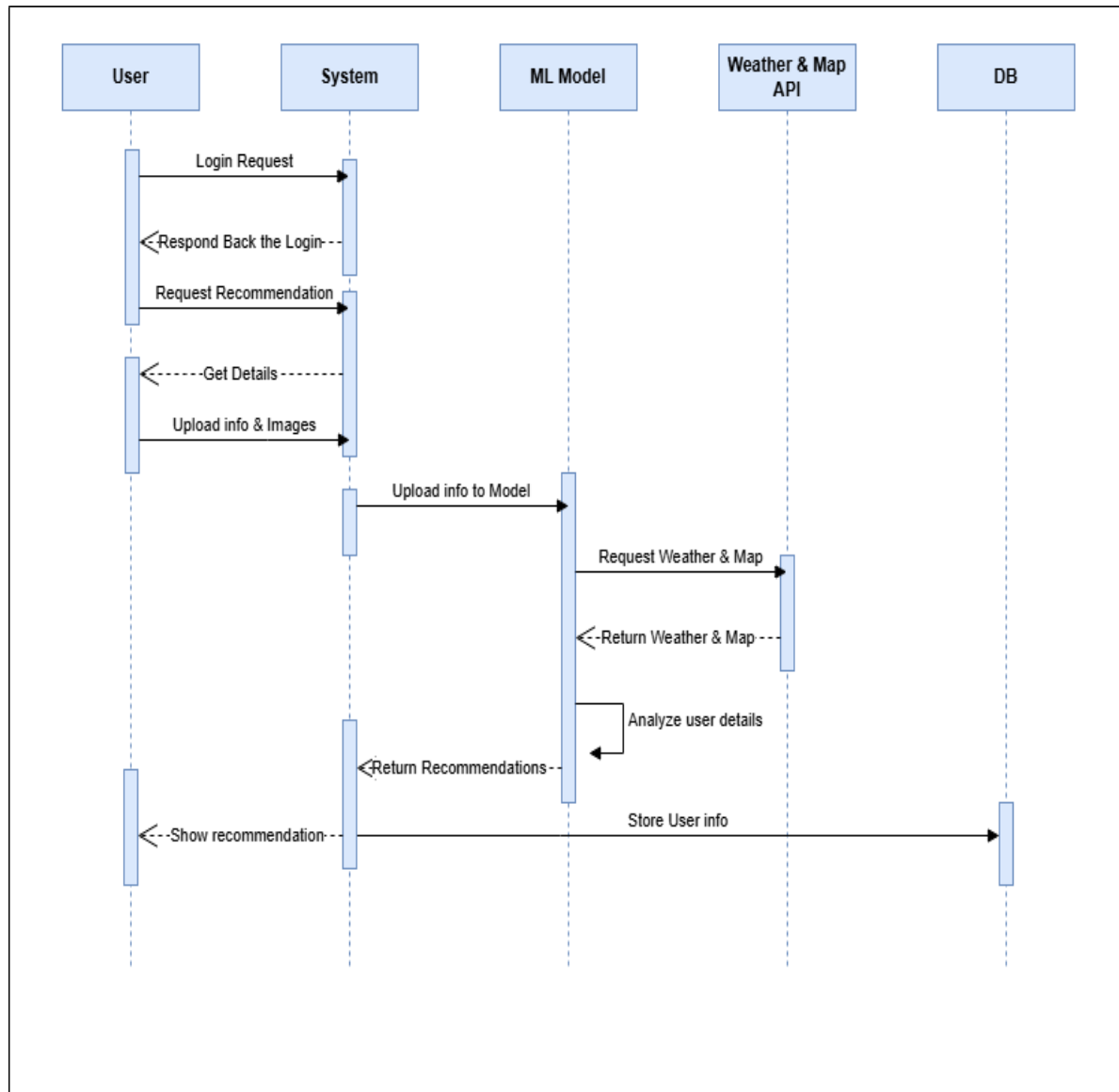
Level 0 DFD



Level 1 DFD



Sequence Diagrams



Class Diagram

MrDresser

